

Workload Picked
Batch of requests through various Engines
LLM Request Timelines
NSYS Measurements
Throughput vs Load for vLLM
GPU Memory Breakdown
Future Steps

Benchmarking and Performance Analysis of LLM Inferencing Systems

Systems for LLMs
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Workload Picked



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- Targeted a very specific workload
- Large system prompt ($\approx$ 1500 tokens) containing useful context and shared across all the queries
- Varying query input rate (from 5 req/sec to 700 req/sec)
- Small query sizes (10 to 20 tokens each) and small output requirements (capped at 30 to 50 tokens)
- Ideally, prefix sharing should help a lot
- **Next Step:** Family of requests, each family with a particular large system prompt and a few queries

Batch of requests through various Engines



Technique Used	**Time (s)**
Transformers (batch size = 1)	30.21
Transformers (batch size = 2)	19.51
Transformers (batch size = 3)	18.64
vLLM (no system prompt pre-filling)	0.45
SGLang (fork)	1.03
SGLang (run_batch)	0.40
NEO	0.55

Table: Measuring the time taken to process a batch of 24 requests using different techniques

Batch of requests through various Engines

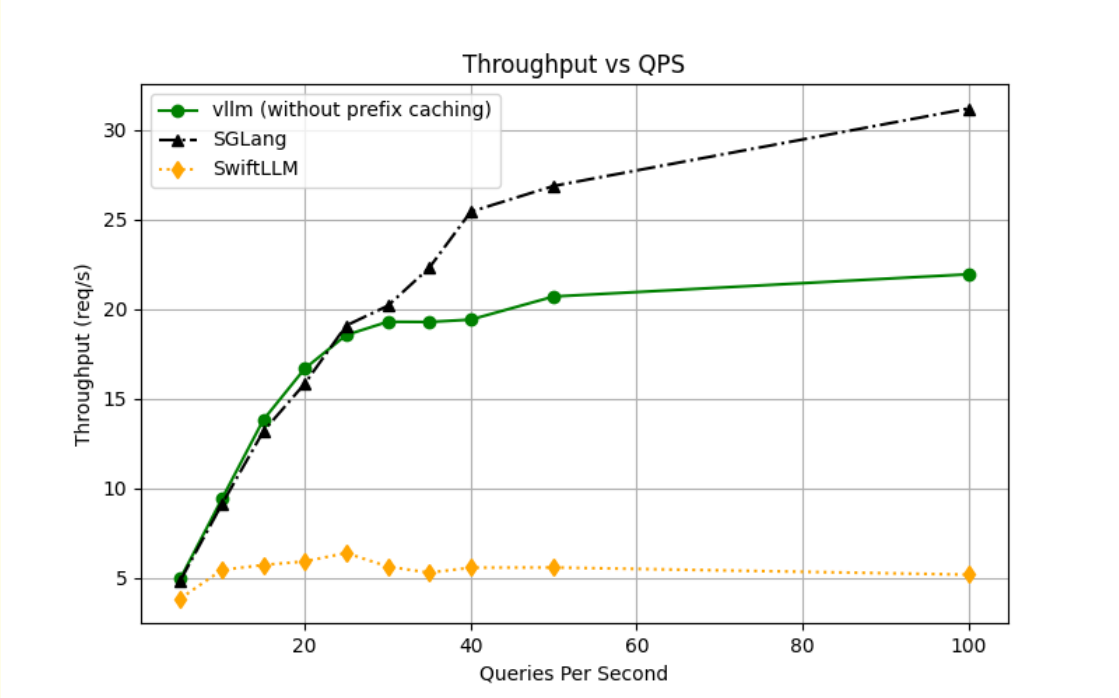


Figure: Throughput comparison across varying workloads

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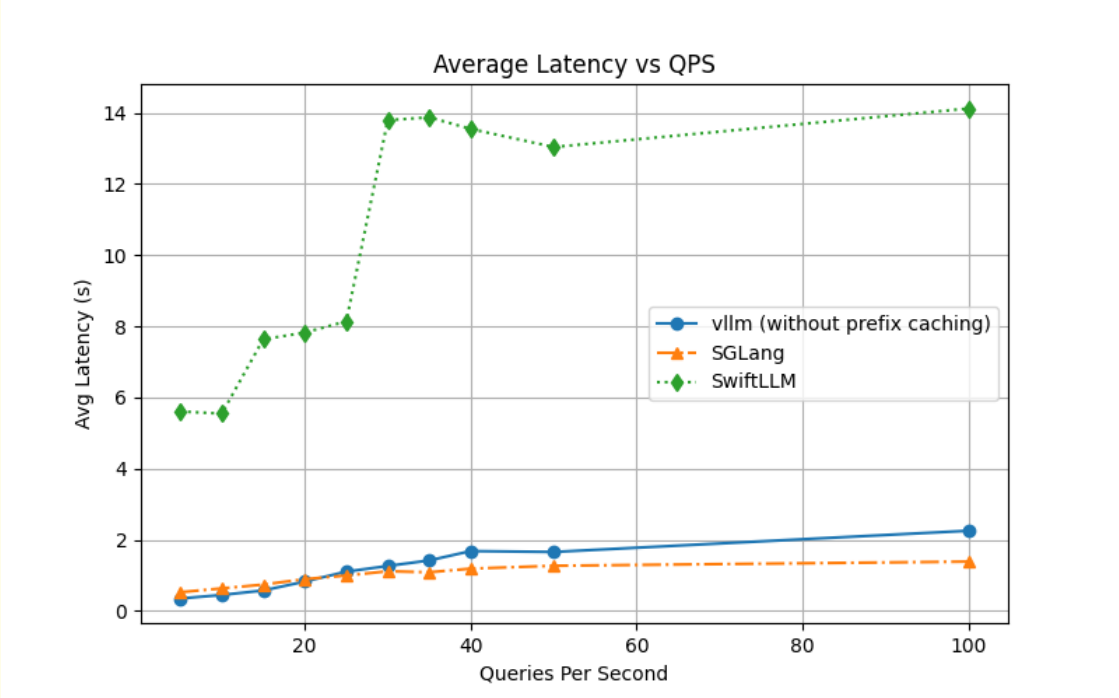


Figure: Average Latency comparison across varying workloads

LLM Request Timelines

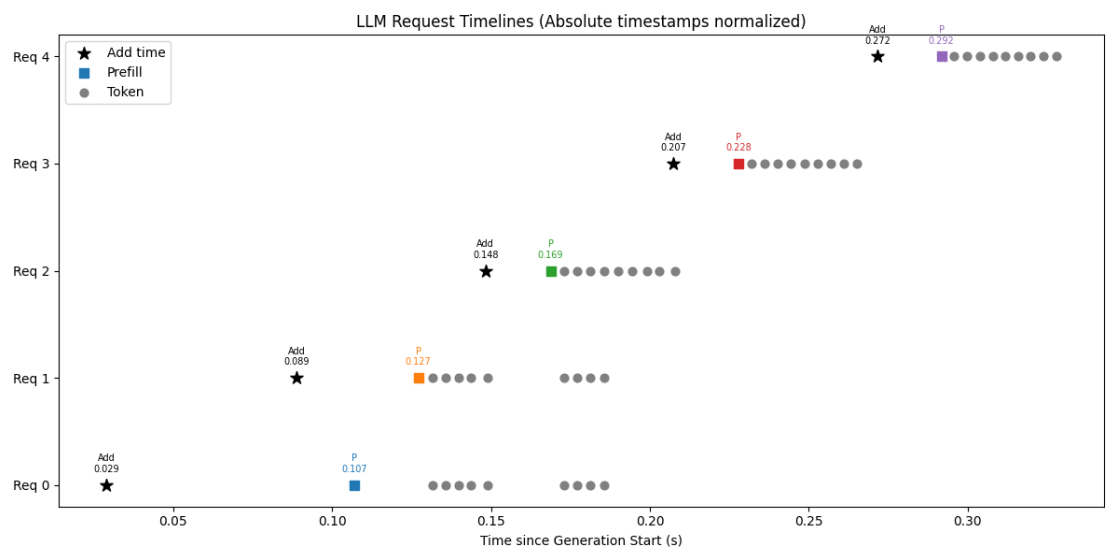


Figure: 20 Requests per second - without chunked prefill - vLLM V0 Engine

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NSYS Measurements



Figure: NSYS Metrics Snapshot

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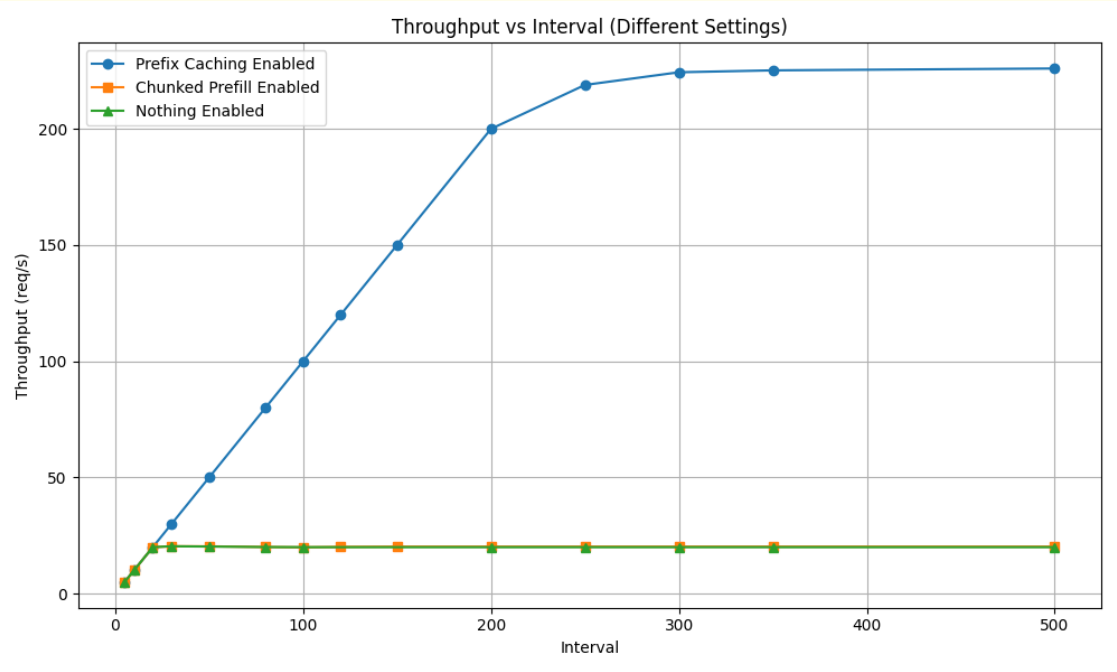


Figure: Throughput vs Load

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GPU Memory Breakdown



Component	Memory Occupied
Weights	0.93 GiB
Peak Activation	4.07 GiB
Non-Torch	0.02 GiB
CUDA Graph	0.41 GiB
KV Cache: Intersection (Prefix)	1,280 tokens / 0.0157 GiB
KV Cache: Extras for Requests	640 tokens (10 requests × 64 tokens) / 0.0078 GiB
KV Cache: Free	2,672,880 tokens / 32.8445 GiB

Table: GPU Memory Breakdown and KV Cache Usage

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- Find the max throughput we can achieve for this kind of workload through vLLM
- Find the throughput for a family of requests, each family sharing a system prompt amongst its members
- Explore other batching techniques for such a family of requests, taking prefix sharing into consideration
- Possible changes to the scheduling algorithm
- Can we offload stuff to CPU, when we have computed the KV Cache, and we just need to perform decode operations?
- Possible optimizations to fine tuning



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Thank You